

GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built self-service tooling and reusable Python SDKs that let internal teams create and manage their own data/release pipelines without needing to understand the platform's internals underneath.
- Built data quality and observability tooling: automated data/drift checks, dataset validation, and usage auditing across pipelines, blocking 15% of releases that failed validation before reaching production.
- Debugged production issues across services and infrastructure layers; built observability and autoscaling with Prometheus and Grafana on AWS (Lambda, S3, DynamoDB) to keep the platform stable through node loss.

QureAI

Jan 2024 – May 2025

ML Infrastructure Engineer

- Built PySpark data processing jobs for large-scale feature/dataset preparation, and led migration of a synchronous service to an async, autoscaled architecture, cutting p99 latency 420ms to 140ms across 50K+ daily requests.
- Built a production observability and event-logging framework tracking latency, drift, and regressions across service versions to keep large-scale data systems reliable.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and Java microservices in a distributed, fault-tolerant system for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS) with relational and NoSQL data stores.
- Drove a redesign of nightly financial data aggregation and batch processing across large-scale distributed data stores, cutting processing from 8 minutes to 90 seconds via indexed views and partitioning.
- Mentored engineers and led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams, reducing production escalations 60% in one quarter through architecture reviews.

Projects

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production performance and eval tooling with rubric-based scoring, drift detection, and per-call latency benchmarking across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, SQL, Go, C++

Data Systems: Data Pipelines, ETL/ELT, PySpark, Data Quality & Observability, Dataset Validation, Airflow

Distributed Systems: Fault Tolerance, High Availability, Autoscaling, Prometheus, Grafana

Databases: PostgreSQL, DynamoDB, MongoDB, Relational & NoSQL Design

Cloud & Infra: AWS (EC2, Lambda, S3, DynamoDB, API Gateway), GCP (Cloud Run), Docker, Kubernetes, Git, Linux, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer